Duc Toan Nguyen

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Education

Rice University, Houston, TX — Aug 2025 – Present
Ph.D. in Electrical and Computer Engineering

Texas Christian University, Fort Worth, TX — May 2025
B.S. in Mathematics; B.S. in Computer Science
John V. Roach Honors College

Industry Experience

KLA Corporation, Ann Arbor, MI — May 2026 – Aug 2026
Algorithm Engineer Intern

- Developed a **deep learning** pipeline for robust semiconductor reticle inspection under strict **runtime and memory constraints**, improving applicability to real-time industrial systems.
- Designed a novel anomaly-scoring method to localize sparse misalignment regions in high-resolution optical images and support reliable defect detection.
- Evaluated state-of-the-art representation and vision methods including persistent homology and Morse-Smale graph representations.

Research Experience

Rice University — Feb 2026 – Present
Graduate Researcher, Decentralized and Multi-Agent Optimization

- Develop an **intrinsic decentralized optimization framework** for networks of interacting agents, achieving exact convergence in both **agent consensus and global objective optimization**.
- Design mathematically rigorous coordination algorithms for **distributed and multi-agent systems** under deterministic and stochastic information, with analysis of convergence, robustness, and network effects.
- Establish scalable algorithms across general nonlinear state spaces including sphere, Grassmann, Hadamard, and Bures–Wasserstein geometries.

Rice University — Aug 2025 – Present
Graduate Researcher, Fréchet Regression and Geometric Machine Learning

- Develop scalable **geometric machine learning and optimization** methods for structured matrix-valued data using Bures–Wasserstein geometry.
- Design and analyze algorithms for regression and learning on graph Laplacians and covariance matrices, with applications to network data and diffusion tensor MRI.

Independent Research — Apr 2024 - May 2025
Researcher, Scalable Stochastic Machine Learning

- Developed a novel **variance-reduced stochastic optimization** algorithm combining gradient variance reduction with data shuffling for **large-scale machine learning**.
- Proved improved convergence complexity relative to existing methods, addressing **computational efficiency and scalable model training**.
- Developed an inexact algorithm for Expectation Maximization problems with rigorous computational-complexity guarantees.

Texas Christian University — Jan 2023 – May 2025
Research Assistant, GO2AI Project

- Developed decision-making architectures for distributed **AI agents** by integrating **Monte Carlo Tree Search (MCTS)**, convolutional neural networks, and reinforcement learning for autonomous Go agents.
- Improved **agent planning and policy learning** through distributed training methods and experimental evaluation of learning behavior.
- Implemented **Grad-CAM explainability** to interpret black-box agent decisions and analyze learned policies, supporting more **transparent and trustworthy AI systems**.

Rice University, Department of Statistics — May 2023 – Jul 2023
Undergraduate Researcher, STAT-DATASCI REU

- Developed "Square-Root Min-Vol NMF," a new representation-learning algorithm with rigorous convergence guarantees for high-dimensional data.
- Conducted extensive experiments on hyperspectral datasets and demonstrated improved reconstruction performance over recent methods.

Academic Publications

- **Nguyen, D. T.**, & Uribe, C. A. "Fréchet Regression on the Bures-Wasserstein Manifold." Neural Information Processing Systems (NeurIPS) 2026.
- **Nguyen, D. T.**, & Uribe, C. A. "On the Finite-Time Analysis for Intrinsic Decentralized Riemannian Optimization." NeurIPS Optimization for Machine Learning (OPT) Workshop 2026.
- **Nguyen, D. T.**, & Uribe, C. A. "Intrinsic Decentralized Stochastic Riemannian Optimization on Manifolds with Bounded Sectional Curvature." IEEE Control Systems Letters (2026). Invited for presentation at the *IEEE Conference on Decision and Control (CDC 2026)*.
- **Nguyen, D. T.**, Tran, T. H., & Nguyen, L. M. "Adjusted Shuffling SARAH: Advancing Complexity Analysis via Dynamic Gradient Weighting". Journal of Optimization Theory and Applications, 210(1), (2026): 12.
- **Nguyen, D. T.** "On the existence of a balanced vertex in geodesic nets with three boundary vertices." *Journal of Geometry*, 116.3 (2025): 36.
- **Nguyen, D. T.**, & Chi, E. C. "Towards tuning-free minimum-volume nonnegative matrix factorization." *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM24)*.
- **Nguyen, D. T.** "Geodesic Nets - Construction and Existence." *(Outstanding Honors Thesis)*.
- Zhou, Y.*, **Nguyen, D. T.***, & Uribe, C. A. "VERI-FRIDA: Geometry-Aware Implicit Variance Reduction for Fréchet Regression." (Submitted to ICLR 2027)

Technical Skills

Programming	Python, C++, C, Java, MATLAB, R, SQL, JavaScript
AI / ML	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Captum, CNNs, Reinforcement Learning, Monte Carlo Tree Search, Explainable AI
Research Areas	AI Agents, Multi-Agent Systems, AI Reliability, Trustworthy AI, Model Evaluation, Distributed Learning, Stochastic Optimization, Scalable ML, Geometric Learning
Systems	Linux, Distributed Computing, Git

Honors & Awards

2026 **Student Travel Funding** – IEEE Control System Society (CSS) for IEEE CDC 2026
2026 **Pi Mu Epsilon Outstanding Speaker Award** – Joint Mathematics Meetings (JMM)
2025 **Senior Scholar** – Department of Mathematics, TCU
2025 **Best Undergraduate Poster** – TCU Student Research Symposium
2025 Finalist, **Best Honors Thesis Presentation** (Boller Competition) – TCU
2025 **Student Travel Funding** – Pi Mu Epsilon (PME) for JMM 2025
2024 **Top 300** (of 3,988 students) – 85th William Lowell Putnam Mathematical Competition
2024 **SIAM Student Travel Award** – SIAM Int. Conference on Data Mining (SDM24)
2023 **Outstanding Session Presentation** – Comp. Math & Operations Research, GCURS
2022 **Top 500** (of 3,415 students) – 83rd William Lowell Putnam Mathematical Competition

Professional Activities

- **Reviewer**, *International Conference on Learning Representations (ICLR) 2026.*
- **Reviewer**, *The Conference on Neural Information Processing Systems (NeurIPS) 2026.*
- **Reviewer**, *Journal of Machine Learning Research (JMLR).*
- **Reviewer**, *Journal of Optimization Theory and Applications (JOTA).*
- **Reviewer**, *Statistica Sinica.*
- **Reviewer**, *IEEE Transactions on Signal and Information Processing over Networks.*
- **Reviewer**, *Computational Optimization and Applications.*
- **Reviewer**, *Discrete & Computational Geometry.*